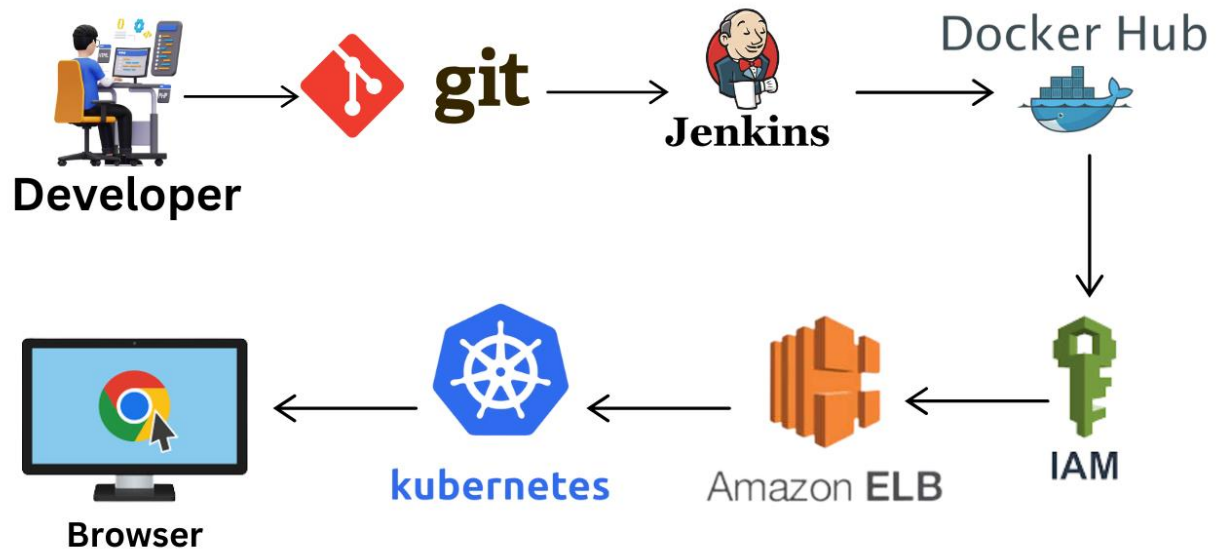


Kubernetes Project Deployment Guide

Architecture:



This document is a step-by-step guide to setting up a **Kubernetes project on AWS** using tools like **Jenkins, Docker, Kops, and GitHub**. It helps you automate the process of deploying a Python application using a **CI/CD pipeline**.

Here's what it covers:

1. **Setting Up Jenkins** on an Ubuntu EC2 instance to automate tasks.
2. **Creating a GitHub Repository** to store and manage your code.
3. **Building a Jenkins Pipeline** to pull code, build a Docker image, and push it to DockerHub.
4. **Setting Up a Kubernetes Cluster** using Kops on AWS to run your application.
5. **Deploying the Application** to Kubernetes automatically through Jenkins.
6. **Setting Up Webhooks** for automatic deployments whenever you update your code.

This guide is perfect for developers who want to quickly set up and automate their application deployments using modern DevOps tools.

Step 1: Create Ubuntu EC2 & Install the Jenkins Server

AMI: Ubuntu 24.04 AMI

Instance Type: t2.medium

Security Group: ALL TCP

The screenshot shows the AWS Management Console for the ap-south-1 region. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity, and Reservations. The main content area displays the 'Resources' section, which lists various EC2 resources in the Asia Pacific (Mumbai) Region. A table shows the following counts: Instances (running) 0, Auto Scaling Groups 0, Capacity Reservations 0, Dedicated Hosts 0, Elastic IPs 0, Instances 0, Key pairs 3, Load balancers 0, Placement groups 0, Security groups 3, Snapshots 0, and Volumes 0. Below the table, the 'Launch instance' button is highlighted with a red box. The 'Service health' section shows the AWS Health Dashboard and the Region as Asia Pacific (Mumbai). The right sidebar displays the 'EC2 Free Tier' information, including the end of month forecast and the offer usage (monthly) for Linux EC2 Instances.

The screenshot shows the 'Instance type' selection screen in the AWS Management Console. The 'Instance type' dropdown menu is open, and 't2.medium' is selected. The dropdown menu lists the following details for the t2.medium instance type: Family: t2, 2 vCPU, 4 GiB Memory, Current generation: true. Below this, the pricing for different operating systems is listed: On-Demand Linux base pricing: 0.0496 USD per Hour, On-Demand Ubuntu Pro base pricing: 0.0531 USD per Hour, On-Demand Windows base pricing: 0.0676 USD per Hour, On-Demand RHEL base pricing: 0.0784 USD per Hour, and On-Demand SUSE base pricing: 0.1496 USD per Hour. The 'All generations' radio button is selected. A link to 'Compare instance types' is visible on the right.

▼ Configure storage Info

Advanced

1x 30 GiB gp3 Root volume (Not encrypted)

Instances (1) Info

Last updated
less than a minute ago



Connect

Instance state ▼

Actions ▼

Launch instances



Find Instance by attribute or tag (case-sensitive)

All states ▼

Instance ID = i-03f03ba878447aea9

Clear filters

< 1 > ⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	k8s_project_1	i-03f03ba878447aea9	Running	t2.medium	Initializing	View alarms +

Update all software packages on Ubuntu server

```
sudo su - root
sudo apt-get update
sudo apt-get upgrade
```

```
$ sudo apt-get update
[sudo] password for developer:
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
```

```
$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  linux-aws linux-headers-aws linux-image-aws sosreport
The following packages will be upgraded:
  amd64-microcode cloud-init distro-info-data gcc-14-base initramfs-tools initramfs-tools-bin initramfs-tools-core libarchive13t64 libgcc-s1 libproc2-0
  libstdc++6 linux-tools-common login mtr-tiny nano passwd procs python3-update-manager python3-urllib3 snapd update-manager-core vim vim-common
  vim-runtime vim-tiny xxd
26 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
Need to get 44.3 MB of archives.
After this operation, 2094 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 login amd64 1:4.13+dfsg1-4ubuntu3.2 [202 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-14-base amd64 14.2.0-4ubuntu2~24.04 [50.8 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libstdc++6 amd64 14.2.0-4ubuntu2~24.04 [791 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgcc-s1 amd64 14.2.0-4ubuntu2~24.04 [78.6 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 passwd amd64 1:4.13+dfsg1-4ubuntu3.2 [845 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libproc2-0 amd64 2:4.0.4-4ubuntu3.2 [50.5 kB]
```

Install Java on Ubuntu server

```
sudo apt-get install default-jdk
java -version
```

```
$ sudo apt-get install default-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service
default-jdk-headless default-jre default-jre-headless fontconfig fontconfig-config fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-mono
gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common libasound2-data libasound2t64 libatk-bridge2.0-0t64
libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3 libavahi-common-data libavahi-common3 libcairo-gobject2
libcairo2 libcolor2 libcupst264 libdatrie1 libdconf1 libdeflate0 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libepoxy0 libfontconfig1
libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-amd-glx libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0
libglx0 libgraphite2-3 libgtk-3-0t64 libgtk-3-bin libgtk-3-common libharfbuzz0b libice-dev libice6 libjbig0 libjpeg-turbo8 libjpeg8 liblcms2-2 liblerc4
libllvmlt64 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1 libpixman-1-0 libpthread-stubs0-dev librsvg2-2
librsvg2-common libsharpyuv0 libsm-dev libsm6 libthai-data libthai0 libtiff6 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwebp7
libx11-dev libx11-xcb1 libxau-dev libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0
libxcb-shm0 libxcb-sync1 libxcb-xf86-dri3-0 libxcb1-dev libxcomposite1 libxcursor1 libxdamage1 libxdmcp-dev libxfixes3 libxft2 libxi6 libxinerama1
libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt-dev libxt6t64 libxtst6 libxv1 libxxf86dga1 libxxf86vm1 mesa-vulkan-drivers
openjdk-21-jdk openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11-common x11-utils x11proto-dev
xorg-sgml-doctools xtrans-dev
Suggested packages:
alsa-utils libasound2-plugins colord cups-common gvfs libice-doc liblcms2-utils pcsclib librsvg2-bin libsm-doc libx11-doc libxcb-doc libxt-doc
openjdk-21-demo openjdk-21-source visualvm libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic
mesa-utils
Recommended packages:
```

```
$ java -version
openjdk version "21.0.5" 2024-10-15
OpenJDK Runtime Environment (build 21.0.5+11-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 21.0.5+11-Ubuntu-1ubuntu124.04, mixed mode, sharing)
$
```

Jenkins Installation

1. Install Jenkins on Ubuntu using the [official documentation](#).
2. Access Jenkins Url with Public ip on port 8080
3. Retrieve the initial admin password:

```
$cat /var/lib/jenkins/secrets/initialAdminPassword
```

4. Install the Suggested Plugins
5. Setup a UserName and Password

← ↻ 🔍 <https://www.jenkins.io/doc/book/installing/linux/#debianubuntu> 🔍 📄 ⭐ ⚙️

How to install Symf... How to Install Symf... Google mod rewrite - How... nginx CORS on prox... How to enable Cros... enable cross-origi... cors - How do I add... Example Nginx conf...

Jenkins  Blog Success Stories Contributor Spotlight Documentation Plugins Community Subprojects Security About Download 🔍 Search 

> User Documentation Home

User Handbook

- User Handbook Overview
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- ...

Debian/Ubuntu

On Debian and Debian-based distributions like Ubuntu you can install Jenkins through [apt](#).

 With the release of Debian 12, OpenJDK 11 is **no longer** included. It has been replaced with OpenJDK 17, which is reflected in the instructions below.

Long Term Support release

A LTS (Long-Term Support) release is chosen every 12 weeks from the stream of regular releases as the stable release for that time period. It can be installed from the [debian-stable apt repository](#).

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt-get update
sudo apt-get install jenkins
```

Launch an instance | EC2 | ap-south-1 | Instances | EC2 | ap-south-1 | 52.66.209.209

← → × 🔍 52.66.209.209:8080 ☆ 🟢 ⋮



This site can't be reached

52.66.209.209 took too long to respond.

Try:

- Checking the connection
- Checking the proxy and the firewall
- Running Windows Network Diagnostics

ERR_TIMED_OUT

Reload

Details

```
$ sudo su -
root@ip-172-31-43-155:~# sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt-get update
sudo apt-get install jenkins
--2024-11-13 01:23:17-- https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.154.133, 2a04:4e42:24::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.154.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3175 (3.1K) [application/pgp-keys]
Saving to: '/usr/share/keyrings/jenkins-keyring.asc'

/usr/share/keyrings/jenkins-keyring.as 100%[=====>] 3.10K --.-KB/s in 0s

2024-11-13 01:23:17 (49.0 MB/s) - '/usr/share/keyrings/jenkins-keyring.asc' saved [3175/3175]

Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:8 https://pkg.jenkins.io/debian-stable binary/ Packages [28.0 kB]
Fetched 30.9 kB in 0s (66.5 kB/s)
Reading package lists... Done
```

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-059f641bdc7e6d0d1

aws Services Search [Alt+S]

EC2 > **Security Groups** > sg-059f641bdc7e6d0d1 - launch-wizard-1 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sg-07c099e18b486f298	HTTPS	TCP	443	Custom	0.0.0.0/0	Delete
sg-04618342b9d83a44e	SSH	TCP	22	Custom	0.0.0.0/0	Delete
sg-0a0765306ccc0da44	HTTP	TCP	80	Custom	0.0.0.0/0	Delete
-	All TCP	TCP	0 - 65535	Anyw...	0.0.0.0/0	Delete

Sign in [Jenkins]

Not secure 52.66.209.209:8080/login?from=%2F

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

```
root@ip-172-31-43-155:~# cat /var/lib/jenkins/secrets/initialAdminPassword
eabb1f4ee9a54df5a14b156bfc84c4c
root@ip-172-31-43-155:~#
```

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

⚠ Not secure 52.66.209.209:8080

Getting Started

Create First Admin User

Username

alvinov13

Password

.....

Confirm password

.....

Full name

Alvin S

Email address

Jenkins 2.479.1

[Skip and continue as admin](#)

Save and Continue

← → ↻ ⚠ Not secure 52.66.209.209:8080 ☆

 **Jenkins**

   Alvin S

Dashboard >

+ New Item

 Build History

 **Manage Jenkins**

 My Views

Build Queue

▼

No builds in the queue.

Build Executor Status

0/2 ▼

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job +

Set up a distributed build

Set up an agent 

Configure a cloud 

Learn more about distributed builds 

System Configuration

**System**
Configure global settings and paths.

**Tools**
Configure tools, their locations and automatic installers.

**Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

**Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

**Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.

**Appearance**
Configure the look and feel of Jenkins

Security

**Security**
Secure Jenkins; define who is allowed to access/use the system.

**Credentials**
Configure credentials

**Credential Providers**
Configure the credential providers and types

**Users**
Create/delete/modify users that can log in to this Jenkins.

← → ↻ ⚠ Not secure 52.66.209.209:8080/manage/pluginManager/available

 **Jenkins**

🔍 Search (CTRL+K) ⓘ 🛡️ 1 👤 Alvin S ▾ 🚪 log out

Dashboard > Manage Jenkins > Plugins

Plugins

🔍 Search available plugins ⓘ

 Install ⓘ

 Updates

 Available plugins

 Installed plugins

 Advanced settings

 Download progress

Install	Name ↓	Released
☐	JavaMail API 1.6.2-10 Library plugins (for use by other plugins) This plugin provides the JavaMail API for other plugins.	5 mo 23 days ago
☐	Command Agent Launcher 116.vd85919c54a_d6 Agent Management Allows agents to be launched using a specified command.	6 hr 54 min ago
☐	Oracle Java SE Development Kit Installer 80.v8a_dee33ed6f0 Allows the Oracle Java SE Development Kit (JDK) to be installed via download from Oracle's website.	3 mo 8 days ago
☐	Pipeline: REST API 2.34 User Interface Provides a REST API to access pipeline and pipeline run data.	1 yr 0 mo ago
☐	Pipeline: Stage View 2.34 User Interface Pipeline Stage View Plugin.	1 yr 0 mo ago

← → ↻ ⚠ Not secure 52.66.209.209:8080/manage/pluginManager/available

Jenkins

Search (CTRL+K) ? 🔒 1 👤 Alvin S 🚪 log out

Dashboard > Manage Jenkins > Plugins

Plugins

🔍 stage

📦 Install

↻

Install	Name ↓	Released
☒	Pipeline: Stage View 2.34 User Interface Pipeline Stage View Plugin.	1 yr 0 mo ago
☐	Pipeline Aggregator View 104.v94a_e5f6cdb_c3 User Interface Full screen dashboard to view stages of the pipeline as a build history. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	7 mo 18 days ago

🔍 ⚠ Not secure 52.66.209.209:8080/login?from=%2Fmanage%2FpluginManager%2Fupdates%2F



Jenkins is restarting

Your browser will reload automatically when Jenkins is ready

Safe Restart

Builds on agents can usually continue.

Download progress - Plugins [1] x +

← → ↻ ⚠ Not secure 52.66.209.209:8080/manage/pluginManager/updates/

Jenkins

Search (CTRL+K) ? 🔒 1 👤 Alvin S 🚪 log out

Dashboard > Manage Jenkins > Plugins

Plugins

- 📦 Updates
- 📦 Available plugins
- 🔧 Installed plugins
- ⚙ Advanced settings

Download progress

- [Go back to the top page](#)
(you can start using the installed plugins right away)
- ☐ Restart Jenkins when installation is complete and no jobs are running

Step 2: Get to know about Python Code and Create GitHub Repo

https://github.com/new

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Owner *

Alvinbsi

Repository name *

k8s_project_nov13

✓ k8s_project_nov13 is available.

Great repository names are short and memorable. Need inspiration? How about [upgraded-giggle](#) ?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:

☐ Add a README file

This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None

.gitignore template: None

Choose which files not to track from a list of templates. [Learn more about ignoring files.](#)

Choose a license

License: None

A license tells others what they can and can't do with your code. [Learn more about licenses.](#)

 You are creating a public repository in your personal account.

Create repository

https://github.com/Alvinbsi/k8s_project_nov13

k8s_project_nov13 Public

Pin Unwatch 1 Fork 0 Star 0

Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
[Get started with GitHub Copilot](#)

Add collaborators to this repository
Search for people using their GitHub username or email address.
[Invite collaborators](#)

Quick setup — if you've done this kind of thing before

[Set up in Desktop](#) or [HTTPS](#) [SSH](#)

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# k8s_project_nov13" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/Alvinbsi/k8s_project_nov13.git
git push -u origin main
```

> This PC > New Volume (E:) > ALVIN > alvinitworld > PROJECTS > NOV13

Sort View

Name	Date modified	Type	Size
k8s_project_nov13	13-11-2024 07:05 AM	File folder	

MINGW64;e:/ALVIN/alvinitworld/PROJECTS/NOV13

```
$ git clone https://github.com/Alvinbsi/k8s_project_nov13.git
Cloning into 'k8s_project_nov13'...
warning: You appear to have cloned an empty repository.
Alvin Selva@DESKTOP-582MOSL MINGW64 /e:/ALVIN/alvinitworld/PROJECTS/NOV13
$ |
```

```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ vi main.py|
```

vi main.py

```
#uvicorn main:app --reload

from fastapi import FastAPI, Request, Form

from fastapi.templating import Jinja2Templates

app = FastAPI()

templates = Jinja2Templates(directory="/code")

@app.get("/")

def form_post(request: Request):

    return templates.TemplateResponse('form.html',
    context={'request': request})
```

```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ vi pod.yaml|
```

vi pod.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: loadbalancer-svc
spec:
  type: LoadBalancer
  selector:
    app: hello
  ports:
    - name: http
      protocol: TCP
      port: 80
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: loadbalancer-pod
spec:
  replicas: 3
  selector:
    matchLabels:
      app: hello
  template:
    metadata:
      labels:
        app: hello
    spec:
      containers:
        - name: hello
          image: alvinselva/alvink8simage:latest
          imagePullPolicy: Always
```

```

apiVersion: v1
kind: Service
metadata:
  name: loadbalancer-svc
spec:
  type: LoadBalancer
  selector:
    app: hello
  ports:
    - name: http
      protocol: TCP
      port: 80
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: loadbalancer-pod
spec:
  replicas: 3
  selector:
    matchLabels:
      app: hello
  template:
    metadata:
      labels:
        app: hello
    spec:
      containers:
        - name: hello
          image: alvinSelva/alvink8simage:latest
          imagePullPolicy: Always

```

```

Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ vi requirements.txt

```

vi requirements.txt

```

aiofiles==0.5.0
aniso8601==7.0.0
ansible-lint==5.0.7
astroid==2.5.1
async-exit-stack==1.0.1
async-generator==1.10
awscli==1.22.97
boto3==1.21.42
botocore==1.24.42
brace==2.1.1
certifi==2021.5.30
chardet==4.0.0
charset-normalizer==2.0.4
click==7.1.2
colorama==0.4.3
commonmark==0.9.1

```

cryptography==3.4.7
dnspython==2.1.0
docutils==0.15.2
email-validator==1.1.3
enrich==1.2.6
fastapi==0.68.1
graphene==2.1.9
graphql-core==2.3.2
graphql-relay==2.0.1
h11==0.12.0
idna==3.2
isort==5.7.0
itsdangerous==1.1.0
Jinja2==2.11.3
jmespath==0.10.0
lazy-object-proxy==1.5.2
MarkupSafe==2.0.1
mccabe==0.6.1
packaging==20.9
pathspec==0.8.1
pdfminer.six==20211012
promise==2.3
py4j==0.10.9
pyasn1==0.4.8
pycparser==2.20
pydantic==1.8.2
Pygments==2.8.1
pylint==2.7.2
pyparsing==2.4.7
python-dateutil==2.8.1
python-dotenv==0.19.0
python-multipart==0.0.5
requests==2.26.0
rich==10.1.0
rsa==4.5
ruamel.yaml==0.17.4
ruamel.yaml.clib==0.2.2
Rx==1.6.1
s3transfer==0.5.0
six==1.15.0
starlette==0.14.2
tenacity==7.0.0
toml==0.10.2
typing-extensions==3.7.4.3
urllib3==1.26.4
uvicorn==0.13.4
watchgod==0.7
wcmatch==8.1.2
websockets==8.1
wrapt==1.12.1


```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ vi Dockerfile
```

vi Dockerfile

```
FROM python:3.9
WORKDIR /code
COPY ./requirements.txt
/code/requirements.txt
RUN pip install --no-cache-dir --
upgrade -r /code/requirements.txt
COPY ./main.py /code/main.py
COPY ./form.html /code/form.html
CMD ["uvicorn", "main:app", "--host",
"0.0.0.0", "--port", "80"]
```

```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ vi form.html
```

vi form.html

copy the source code from this link:

https://www.w3schools.com/w3css/tryit.asp?filename=tryw3css_templates_band&stacked=h

```
<!DOCTYPE html>
<html>
<head>
<title>W3.CSS Template</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet" href="https://www.w3schools.com/w3css/4/w3.css">
<link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Roboto">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/4.7.0/css/font-awesome.min.css">
<style>
html,body,h1,h2,h3,h4,h5,h6 {font-family: "Roboto", sans-serif}
</style>
</head>
<body class="w3-light-grey">
<!-- Page Container -->
<div class="w3-content w3-margin-top" style="max-width:1400px;">
  <!-- The Grid -->
  <div class="w3-row-padding">
```

Push to files to Github

```
git init
git add .
git commit -m "First commit"
git push
```

```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ git status
On branch master
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        form.html
        main.py
        pod.yaml
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)

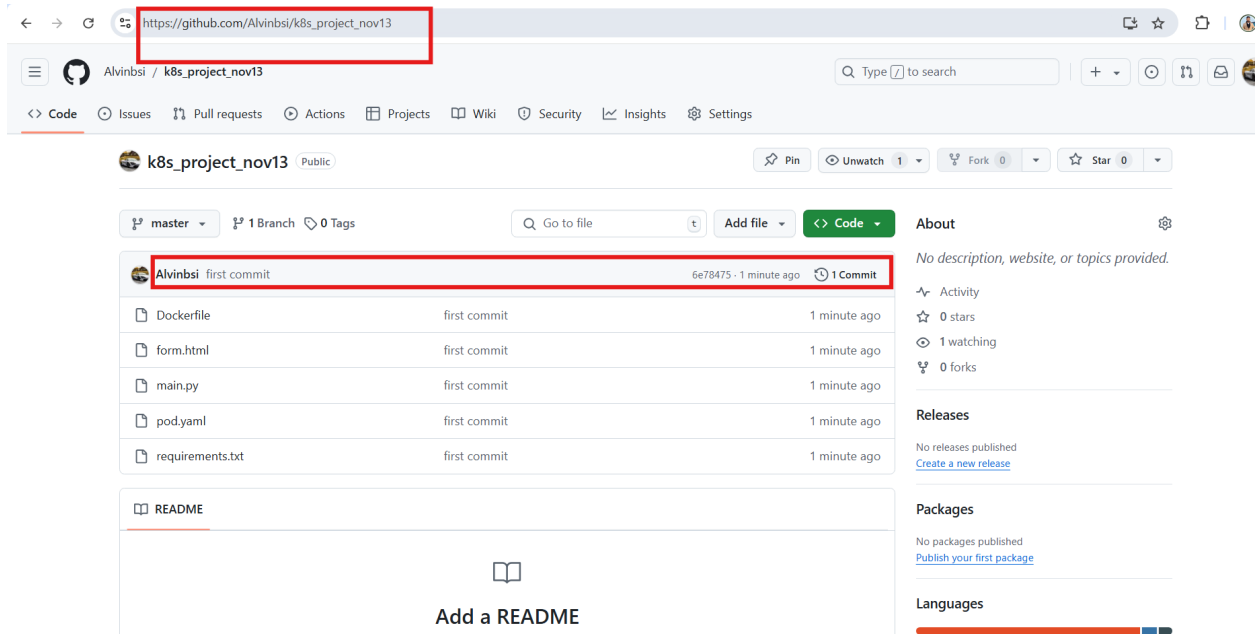
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ git add .
warning: LF will be replaced by CRLF in Dockerfile.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in form.html.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in main.py.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in pod.yaml.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in requirements.txt.
The file will have its original line endings in your working directory

Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ |
```

```
Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ git commit -m "first commit"
[master (root-commit) 6e78475] first commit
 5 files changed, 218 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 form.html
 create mode 100644 main.py
 create mode 100644 pod.yaml
 create mode 100644 requirements.txt

Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.81 KiB | 2.81 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Alvinbsi/k8s_project_nov13.git
 * [new branch]      master -> master

Alvin Selva@DESKTOP-582MOSL MINGW64 /e/ALVIN/alvinitworld/PROJECTS/NOV13/k8s_project_nov13 (master)
$ |
```



Step 3: Create a Jenkins Pipeline Job And clone the Repo

Create a Pipeline Job in v Jenkins

1. Install Stageview plugin from manageJenkins -> Plugins -> Available plugins

New Item

Enter an item name

k8s_build_nov13

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a

OK

Configure

⚙️ General

🔧 Advanced Project Options

🔗 Pipeline

Pipeline

Definition

Pipeline script

Script ?

1

try sample Pipeline...

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save

Apply

1. Use the following pipeline script:

```
pipeline {
  agent any
  stages {
    stage('Pull Code From GitHub') {
      steps {
        git 'https://github.com/Alvinbsi/k8s_project_nov13.git'
      }
    }
  }
}
```

The screenshot shows the Jenkins web interface for configuring a pipeline. The browser address bar shows the URL `52.66.209.209:8080/job/k8s_build_nov13/configure`. The breadcrumb navigation is `Dashboard > k8s_build_nov13 > Configuration`. On the left, the 'Configure' sidebar has three tabs: 'General', 'Advanced Project Options', and 'Pipeline', with 'Pipeline' being the active tab. The main content area is titled 'Pipeline' and shows the 'Definition' dropdown set to 'Pipeline script'. Below this is a 'Script' section with a text area containing the following pipeline script:

```
1 pipeline {
2   agent any
3   stages {
4     stage('Pull Code From GitHub') {
5       steps {
6         git 'https://github.com/Alvinbsi/k8s_project_nov13.git'
7       }
8     }
9   }
10 }
```

Below the script area, there is a checkbox labeled 'Use Groovy Sandbox' which is checked. At the bottom, there are two buttons: 'Save' and 'Apply'.

Configure

General

Advanced Project Options

Pipeline

Pipeline

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent any  
3   stages {  
4     stage('Pull Code From GitHub') {  
5       steps {  
6         git 'https://github.com/Alvinbst/k8s_project_nov13'  
7       }  
8     }  
9   }  
10 }
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save

Apply

← → ↻ ⚠ Not secure 52.66.209.209:8080/job/k8s_build_nov13/

Jenkins 🔍 Search (CTRL+K) 🛡️ 1 👤 Alvi

Dashboard > k8s_build_nov13 >

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

k8s_build_nov13

Stage View

No data available. This Pipeline has not yet run.

Permalinks

Builds

← → ↻ ⚠ Not secure 52.66.209.209:8080/job/k8s_build_nov13/

Jenkins 🔍 Search (CTRL+K)

Dashboard > k8s_build_nov13 >

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

k8s_build_nov13

Stage View

Average stage times:
(Average full run time: ~6s)

#1
Nov 13
07:32
No Changes

Pull Code From
GitHub

4s

4s

Permalinks

Builds

Today

✓ #1 2:02 AM

Step 4: Containerize the Python app and publish to DockerHub

Docker Installation

sudo apt install docker.io

1. Open DockerHub account
docker login

```
ubuntu@ip-172-31-43-155:~$ sudo su -
root@ip-172-31-43-155:~# sudo apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 4 not upgraded.
Need to get 76.8 MB of archives.
After this operation, 289 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
root@ip-172-31-43-155:~# docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/
Username: alvinselva
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store
Login Succeeded
root@ip-172-31-43-155:~#
```

Give Jenkins Permission to run sudo commands

```
root@ip-172-31-43-155:~# vi /etc/sudoers

# User privilege specification
root    ALL=(ALL:ALL) ALL
jenkins ALL=(ALL) NOPASSWD: ALL

# Members of the admin group may gain root privileges
```


← → ↻ 🔍 https://hub.docker.com/repositories/alvinselfva ☆ 📁 🏠

🚀 docker hub Explore Repositories Organizations Usage 🔍 Search Docker Hub **ctrl+K** ⓘ ⚙️ 🔔 🔄 ⋮ A

alvinselfva ▼ 🔍 Search by repository name All Content ▼ [Create repository](#)

alvinselfva / alvin_nginx Contains: Image • Last pushed: 4 months ago	☆ 0	± 5	🔓 Public	🚫 Scout inactive
alvinselfva / httpd_alvin Contains: Image • Last pushed: 4 months ago	☆ 0	± 7	🔓 Public	🚫 Scout inactive
alvinselfva / hello-world-app3 Contains: Image • Last pushed: 6 months ago	☆ 0	± 10	🔓 Public	🚫 Scout inactive

Run the three stages in Jenkinsfile

```
pipeline {
  agent any

  stages {
    stage('Pull Code From GitHub') {
      steps {
        git 'https://github.com/Alvinbsi/k8s_project_nov13.git'
      }
    }
    stage('Build the Docker image') {
      steps {
        sh 'sudo docker build -t alvinimage /var/lib/jenkins/workspace/alvinproj'
        sh 'sudo docker tag alvinimage alvinselfva/alvinimage:latest'
        sh 'sudo docker tag alvinimage alvinselfva/alvinimage:${BUILD_NUMBER}'
      }
    }
    stage('Push the Docker image') {
      steps {
        sh 'sudo docker image push alvinselfva/alvinimage:latest'
        sh 'sudo docker image push alvinselfva/alvinimage:${BUILD_NUMBER}'
      }
    }
  }
}
```

Dashboard > k8s_build_nov13 > Configuration

Configure

General

Advanced Project Options

Pipeline

Pipeline

Definition

Pipeline script

```
Script ?
5 stage('Pull Code From GitHub') {
6   steps {
7     git 'https://github.com/alvinbsi/k8s_project_nov13.git'
8   }
9 }
10 stage('Build the Docker Image') {
11   steps {
12     sh 'sudo docker build -t alvink8simage /var/lib/jenkins/workspace/k8s_project_nov13'
13     sh 'sudo docker tag alvink8simage alvinseiva/alvink8simage:latest'
14     sh 'sudo docker tag alvink8simage alvinseiva/alvink8simage:${BUILD_NUMBER}'
15   }
16 }
17 stage('Push the Docker Image') {
18   steps {
19     sh 'sudo docker image push alvinseiva/alvink8simage:latest'
20     sh 'sudo docker image push alvinseiva/alvink8simage:${BUILD_NUMBER}'
21   }
22 }
23 }
24 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

Dashboard > k8s_build_nov13 >

Jenkins

Search (CTRL+K)

Alvin S

Dashboard > k8s_build_nov13 >

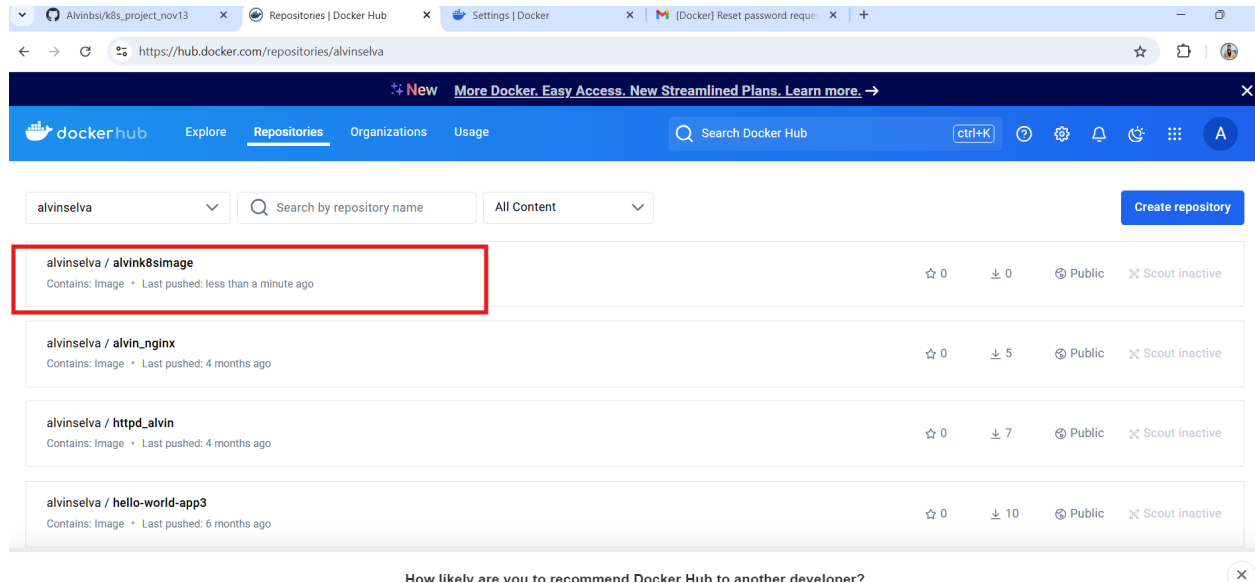
Status

k8s_build_nov13

Add description

Stage View

	Pull Code From GitHub	Build the Docker image	Push the Docker image	Deploy on Kubernetes
Average stage times: (Average full run time: ~26s)	1s	12s	6s	622ms
#6 Nov 13 08:22 No Changes	711ms	919ms	9s	622ms
#7 Nov 13 07:50 No Changes	673ms	1min 1s	25s	



Step 5: Create Kubernetes Cluster

Install Kops

```
curl -Lo kops https://github.com/kubernetes/kops/releases/download/$(curl -s
https://api.github.com/repos/kubernetes/kops/releases/latest | grep tag_name | cut -d '"' -f
4)/kops-linux-amd64
chmod +x ./kops
sudo mv ./kops /usr/local/bin/
```

```
root@ip-172-31-43-155:~# curl -Lo kops https://github.com/kubernetes/kops/releases/download/$(curl -s https://api.github.com/repos/kubernetes/kops/releases
/latest | grep tag_name | cut -d '"' -f 4)/kops-linux-amd64
root@ip-172-31-43-155:~# chmod +x ./kops
root@ip-172-31-43-155:~# sudo mv ./kops /usr/local/bin/
root@ip-172-31-43-155:~#
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left
0	0	0	0	0	0	0	0
100	238M	100	238M	0	0	0	34.9M

Install kubectl

```
curl -Lo kubectl https://storage.googleapis.com/kubernetes-release/release/$(curl -s
https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin/kubectl
```

```
root@ip-172-31-43-155:~# curl -Lo kubectl https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-rel
ease/release/stable.txt)/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin/kubectl
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100 53.7M  100 53.7M    0     0  9.8M      0  0:00:05  0:00:05 --:--:-- 11.2M
root@ip-172-31-43-155:~#
```

AWS CLI Installation

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
sudo apt install unzip
unzip awscliv2.zip
sudo ./aws/install --bin-dir /usr/local/bin --install-dir /usr/local/aws-cli --update
Create IAM user with Admin access policy and generate IAM access & Secret Key
aws configure
aws s3 ls
```

```
root@ip-172-31-43-155:~# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
sudo apt install unzip
unzip awscliv2.zip
sudo ./aws/install --bin-dir /usr/local/bin --install-dir /usr/local/aws-cli --update
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100 63.6M  100 63.6M    0     0 107M      0  --:--:-- --:--:-- --:--:-- 108M
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 4 not upgraded.
Need to get 174 kB of archives.
After this operation, 384 kB of additional disk space will be used.
Get:1 http://ap-south-1-ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd64 6.0-28ubuntu4.1 [174 kB]
Fetched 174 kB in 0s (9273 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 83780 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
```

Create IAM user with Admin access policy and generate IAM access & Secret Key

aws configure

aws s3 ls

The image consists of two screenshots from the AWS IAM console, illustrating the steps to create a new IAM user.

Top Screenshot: Shows the 'Users' page in the IAM console. The breadcrumb navigation 'IAM > Users' is highlighted with a red box. The 'Create user' button is also highlighted with a red box. A green notification banner at the top states 'User "alvink8sIAM" deleted.' The main content area shows 'Users (0)' and a table with columns: User name, Path, Groups, Last activity, MFA, Password age, and Console last. The table is currently empty, displaying 'No resources to display'.

Bottom Screenshot: Shows the 'Specify user details' step of the 'Create user' wizard. The breadcrumb navigation 'IAM > Users > Create user' is visible. The 'User details' section contains a 'User name' field with the text 'k8sIAM' entered. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , _ @ - (hyphen)'. There is an unchecked checkbox labeled 'Provide user access to the AWS Management Console - optional' with a sub-note: 'If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.' A blue information box at the bottom states: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. At the bottom right, there are 'Cancel' and 'Next' buttons.

Step 2
Set permissions

Step 3
Review and create

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1253)

Choose one or more policies to attach to your new user.

Filter by Type
All types

Policy name	Type	Attached entities
☐ AccessAnalyzerServiceRolePolicy	AWS managed	0
☒ AdministratorAccess	AWS managed - job function	1
☐ AdministratorAccess-Amplify	AWS managed	0
☐ AdministratorAccess-AWSElasticB...	AWS managed	0

► Set permissions boundary - optional

Cancel Previous **Next**

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Previous **Create user**

us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/details/k8slAM?section=permissions

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings

Access reports

- Access Analyzer
- External access
- Unused access
- Analyzer settings

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

k8slAM Info Delete

Summary

ARN arn:aws:iam::590183879227:user/k8slAM	Console access Disabled	Access key 1 Create access key
Created November 13, 2024, 08:00 (UTC+05:30)	Last console sign-in -	

Permissions Groups Tags Security credentials Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

Remove Add permissions

Access key best practices & alternatives Info

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

Command Line Interface (CLI)

You plan to use this access key to enable the AWS CLI to access your AWS account.

us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/details/k8slAM/create-access-key

Services Search [Alt+S]

IAM > Users > k8slAM > Create access key

Step 1
[Access key best practices & alternatives](#)

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Set description tag - optional Info

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value

Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidentially later.

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / = + - @

Cancel Previous **Create access key**

aws Services Search [Alt+S]

☑ Access key created
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Step 2 - optional
[Set description tag](#)

Step 3
Retrieve access keys

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
 AKIAYS2NTKI56ZVRMQKR	 ***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) [Done](#)

```
last login: Wed Nov 13 02:18:41 2024 from 152.58.199.253
ubuntu@ip-172-31-43-155:~$ sudo su -
root@ip-172-31-43-155:~# aws configure
AWS Access Key ID [None]: AKIAYS2NTKI56ZVRMQKR
AWS Secret Access Key [None]: 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
default region name [None]: ap-south-1
default output format [None]: json
root@ip-172-31-43-155:~#
```


← → ↻ ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1&bucketType=general

aws Services 🔍 Search [Alt+S]

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Asia Pacific (Mumbai) ap-south-1

Bucket name [Info](#)
k8sbucketnov13

Bucket name must be unique within the global namespace and follow the bucket naming rule. [See rules for bucket naming](#)

Copy settings from existing bucket - *optional*
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

- ☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
- ☐ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
 - ☐ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
 - ☐ **Block public access to buckets and objects granted through *new* public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
 - ☐ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.



Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

- ☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Cancel

► Advanced settings

i After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

Set Env Variables for Kops

```
export AWS_ACCESS_KEY_ID=  
export AWS_SECRET_ACCESS_KEY=  
export NAME=**alvin**.k8s.local  
export KOPS_STATE_STORE=s3://<bucketname>
```

Create Kubernetes Cluster

```
kops create cluster --zones ap-south-1a ${NAME}  
kops update cluster --name alvin.k8s.local --yes --admin
```

```
2024-11-12 07:55:43 alvin@k8sbucket  
root@ip-172-31-43-155:~# export AWS_ACCESS_KEY_ID=AKIAYS2NTKI56ZVRMOKR  
root@ip-172-31-43-155:~# export AWS_SECRET_ACCESS_KEY=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  
root@ip-172-31-43-155:~# export NAME=alvin.k8s.local  
root@ip-172-31-43-155:~# export KOPS_STATE_STORE=s3://k8sbucketnov13  
root@ip-172-31-43-155:~# kops create cluster --zones ap-south-1a ${NAME}  
I1113 02:40:09.764311 12659 new_cluster.go:1435] Gossip is deprecated, using None DNS instead  
I1113 02:40:09.764369 12659 new_cluster.go:1454] Cloud Provider ID: "aws"  
I1113 02:40:11.046573 12659 subnets.go:224] Assigned CIDR 172.20.0.0/16 to subnet ap-south-1a  
Previewing changes that will be made:  
I1113 02:40:17.428427 12659 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/amd64/kubelet" is not well-known, downloading hash  
I1113 02:40:17.755450 12659 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/amd64/kubect1" is not well-known, downloading hash  
I1113 02:40:18.121745 12659 builder.go:312] asset "https://github.com/containerd/containerd/releases/download/v1.7.16/containerd-1.7.16-linux-amd64.tar  
z" is not well-known, downloading hash  
I1113 02:40:20.045927 12659 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/amd64/nodeup" is not well-known, downloading has  
I1113 02:40:20.351961 12659 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/arm64/kubelet" is not well-known, downloading hash  
I1113 02:40:20.867185 12659 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/arm64/kubect1" is not well-known, downloading hash  
I1113 02:40:21.245238 12659 builder.go:312] asset "https://github.com/containerd/containerd/releases/download/v1.7.16/containerd-1.7.16-linux-arm64.tar  
z" is not well-known, downloading hash  
I1113 02:40:23.402281 12659 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/arm64/nodeup" is not well-known, downloading has
```

```
AutoscalingGroup      name:nodes-ap-south-1a,alvin.k8s.local  id:nodes-ap-south-1a,alvin.k8s.local

Must specify --yes to apply changes

Cluster configuration has been created.

Suggestions:
* list clusters with: kops get cluster
* edit this cluster with: kops edit cluster alvin.k8s.local
* edit your node instance group: kops edit ig --name=alvin.k8s.local nodes-ap-south-1a
* edit your control-plane instance group: kops edit ig --name=alvin.k8s.local control-plane-ap-south-1a

Finally configure your cluster with: kops update cluster --name alvin.k8s.local --yes --admin

root@ip-172-31-43-155:~#
```

```
root@ip-172-31-43-155:~# kops update cluster --name alvin.k8s.local --yes --admin
I1113 02:41:42.845466 12686 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/amd64/kubelet" is not well-known, downloading hash
I1113 02:41:43.167792 12686 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/amd64/kubectcl" is not well-known, downloading hash
I1113 02:41:43.544401 12686 builder.go:312] asset "https://github.com/containerd/containerd/releases/download/v1.7.16/containerd-1.7.16-linux-amd64.tar.g
z" is not well-known, downloading hash
I1113 02:41:45.214332 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/amd64/nodeup" is not well-known, downloading hash
I1113 02:41:45.381600 12686 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/arm64/kubelet" is not well-known, downloading hash
I1113 02:41:45.682955 12686 builder.go:312] asset "https://dl.k8s.io/release/v1.30.2/bin/linux/arm64/kubectcl" is not well-known, downloading hash
I1113 02:41:46.048591 12686 builder.go:312] asset "https://github.com/containerd/containerd/releases/download/v1.7.16/containerd-1.7.16-linux-arm64.tar.g
z" is not well-known, downloading hash
I1113 02:41:47.641296 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/arm64/nodeup" is not well-known, downloading hash
I1113 02:41:47.720101 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/amd64/protokube" is not well-known, downloading ha
sh
I1113 02:41:47.795601 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/arm64/protokube" is not well-known, downloading ha
sh
I1113 02:41:47.851991 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/amd64/channels" is not well-known, downloading has
h
I1113 02:41:47.938743 12686 builder.go:312] asset "https://artifacts.k8s.io/binaries/kops/1.30.1/linux/arm64/channels" is not well-known, downloading has
```

```
root@ip-172-31-43-155:~# kubectl get pods
No resources found in default namespace.
root@ip-172-31-43-155:~#
```

Step 6: Deploy to kubernetes

Add a new stage in Jenkins file

```
pipeline {
  agent any

  stages {
    stage('Pull Code From GitHub') {
      steps {
        git 'https://github.com/Alvinbsi/k8s_project_nov13.git'
      }
    }
    stage('Build the Docker image') {
      steps {
        sh 'sudo docker build -t alvinimage /var/lib/jenkins/workspace/alvinproj'
        sh 'sudo docker tag alvinimage alvinselva/alvinimage:latest'
        sh 'sudo docker tag alvinimage alvinselva/alvinimage:${BUILD_NUMBER}'
      }
    }
    stage('Push the Docker image') {
      steps {
        sh 'sudo docker image push alvinselva/alvinimage:latest'
        sh 'sudo docker image push alvinselva /alvinimage:${BUILD_NUMBER}'
      }
    }
    stage('Deploy on Kubernetes') {
      steps {
        sh 'sudo kubectl apply -f /var/lib/jenkins/workspace/alvinproj/pod.yaml'
        sh 'sudo kubectl rollout restart deployment loadbalancer-pod'
      }
    }
  }
}
```

Dashboard > k8s_build_nov13 > Configuration

Configure

Definition: Pipeline script

Script

```
11 steps {
12   sh 'sudo docker build -t alvink8sinage /var/lib/jenkins/workspace/k8s_build_nov13'
13   sh 'sudo docker tag alvink8sinage alvinselva/alvink8sinage:latest'
14   sh 'sudo docker tag alvink8sinage alvinselva/alvink8sinage:${BUILD_NUMBER}'
15 }
16
17 stage('Push the Docker image') {
18   steps {
19     sh 'sudo docker image push alvinselva/alvink8sinage:latest'
20     sh 'sudo docker image push alvinselva/alvink8sinage:${BUILD_NUMBER}'
21   }
22 }
23
24 stage('Deploy on Kubernetes') {
25   steps {
26     sh 'sudo kubectl apply -f /var/lib/jenkins/workspace/k8s_build_nov13/pod.yaml'
27     sh 'sudo kubectl rollout restart deployment loadbalancer-pod'
28   }
29 }
30 }
```

☒ Use Groovy Sandbox

Pipeline Syntax

Save Apply

Dashboard > k8s_build_nov13

Jenkins

Search (CTRL+K)

Status **k8s_build_nov13** Add descrip

<> Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

GitHub Hook Log

Stage View

	Pull Code From GitHub	Build the Docker image	Push the Docker image	Deploy on Kubernetes
Average stage times: (Average full run time: ~26s)	1s	12s	6s	622ms
#6 Nov 13 08:22 No Changes	711ms	919ms	9s	622ms
#7 Nov 13 07:50 No Changes	673ms	1min 1s	25s	

```
root@ip-172-31-43-155:~# kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
loadbalancer-pod-5c8cd59ff-5gsrw    0/1     ContainerCreating   0           68s
loadbalancer-pod-8698564578-4fc7v   0/1     ContainerCreating   0           68s
loadbalancer-pod-8698564578-rgrvd   0/1     ContainerCreating   0           68s
loadbalancer-pod-8698564578-spxvz   0/1     ContainerCreating   0           68s
root@ip-172-31-43-155:~#
```

Load balancers (2)
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	DNS name	State	VPC ID	Availability Zones	Type
aee849e130b44403da...	aee849e130b44403da850...	-	vpc-0b6f3578bb6f011...	ap-south-1a (aps1-az1)	classic
api-alvin-k8s-local-sa6egj...	api-alvin-k8s-local-sa6egj...	Active	vpc-0b6f3578bb6f011...	ap-south-1a (aps1-az1)	network

0 load balancers selected
Select a load balancer above.

Configure Webhook in github & Jenkins

Github Repo Setting -> Webhooks -> http://<publicip>:8080/github-webhook/

Jenkins Job -> Build Triggers -> GitHub hook trigger for GITScm polling

https://github.com/Alvinbsi/k8s_project_nov13

Alvinbsi / k8s_project_nov13

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

k8s_project_nov13 Public

master 1 Branch 0 Tags

Go to file Add file Code

Alvinbsi first commit 6e78475 · 53 minutes ago 1 Commit

File	Commit	Time
Dockerfile	first commit	53 minutes ago
form.html	first commit	53 minutes ago
main.py	first commit	53 minutes ago
pod.yaml	first commit	53 minutes ago
requirements.txt	first commit	53 minutes ago

README

Add a README

About: No description, website, or topics provided.

Activity: 0 stars, 1 watching, 0 forks

Releases: No releases published. Create a new release

Packages: No packages published. Publish your first package

Languages:

← → ↻ https://github.com/Alvinbsi/k8s_project_nov13/settings    

Alvinbsi / k8s_project_nov13    

<> Code Issues Pull requests Actions Projects Wiki Security Insights **Settings**

 **General**

General

Access

- Collaborators
- Moderation options

Code and automation

- Branches
- Tags
- Rules
- Actions
- Webhooks**
- Environments
- Codespaces
- Pages

Security

- Code security
- Deploy keys

Repository name

k8s_project_nov13

Rename

☐ **Template repository**

Template repositories let users generate new repositories with the same directory structure and files. [Learn more about template repositories.](#)

☐ **Require contributors to sign off on web-based commits**

Enabling this setting will require contributors to sign off on commits made through GitHub's web interface. Signing off is a way for contributors to affirm that their commit complies with the repository's terms, commonly the [Developer Certificate of Origin \(DCO\)](#). [Learn more about signing off on commits.](#)

Default branch

The default branch is considered the "base" branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.

master 

Social preview

Upload an image to customize your repository's social media preview.

← → ↻ https://github.com/Alvinbsi/k8s_project_nov13/settings/hooks    

Alvinbsi / k8s_project_nov13   

<> Code Issues Pull requests Actions Projects Wiki Security Insights **Settings**

 **General**

Webhooks

Access

- Collaborators
- Moderation options

Code and automation

- Branches
- Tags
- Rules
- Actions
- Webhooks**
- Environments
- Codespaces
- Pages

Security

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Webhooks / Add webhook

We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, `x-www-form-urlencoded`, etc). More information can be found in [our developer documentation](#).

Payload URL *

Content type *

Secret

SSL verification

By default, we verify SSL certificates when delivering payloads.

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ <http://52.66.209.209:8080/github-w...> (push)

Last delivery was successful.

Edit

Delete

← → ↻ ⚠ Not secure 52.66.209.209:8080/job/k8s_build_nov13/configure ☆ Incognito

Jenkins 🔍 Search (CTRL+K) 🛡 1 👤 Alvin S 🚪 log out

Dashboard > k8s_build_nov13 > Configuration

Configure

- General**
- Advanced Project Options
- Pipeline

General

Enabled

Description

Plain text [Preview](#)

- ☐ Discard old builds ?
- ☐ Do not allow concurrent builds
- ☐ Do not allow the pipeline to resume if the controller restarts
- ☐ GitHub project
- ☐ Pipeline speed/durability override ?

[Preserve stashes from completed builds ?](#)

[Save](#) [Apply](#)

← → ↻ ⚠ Not secure 52.66.209.209:8080/job/k8s_build_nov13/configure

Dashboard > k8s_build_nov13 > Configuration

Configure

- General**
- Advanced Project Options
- Pipeline

- ☐ Do not allow concurrent builds
- ☐ Do not allow the pipeline to resume if the controller restarts
- ☐ GitHub project
- ☐ Pipeline speed/durability override ?
- ☐ Preserve stashes from completed builds ?
- ☐ This project is parameterized ?
- ☐ Throttle builds ?

Build Triggers

- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☒ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?
- ☐ Quiet period ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

[Save](#) [Apply](#)

Make a Commit & Run the Deployment

The screenshot shows the Visual Studio Code editor interface. The Explorer pane on the left shows the project structure for 'K8S_PROJECT_NOV13', including files like Dockerfile, form.html, main.py, pod.yaml, requirements.txt, and Jenkinsfile. The main editor area displays the 'pod.yaml' file with the following content:

```
18 spec:
19   selector:
20     matchLabels:
21       app: hello
22   template:
23     metadata:
24       labels:
25         app: hello
26   spec:
27     containers:
28       - name: hello
29         image: alvinselve/alvink8simage:latest
30         imagePullPolicy: Always
```

The terminal window at the bottom shows the output of a 'git push' command, indicating a successful push to the 'master' branch. The terminal text is highlighted with a red box:

```
create mode 100644 Jenkinsfile
PS E:\ALVIN\alvinitworld\PROJECTS\NOV13\k8s_project_nov13> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 606 bytes | 606.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Alvinbsi/k8s_project_nov13.git
6e78475..2e984d0 master -> master
PS E:\ALVIN\alvinitworld\PROJECTS\NOV13\k8s_project_nov13>
```

The screenshot shows the GitHub repository page for 'Alvinbsi / k8s_project_nov13'. The repository is public and has 1 branch (master) and 0 tags. The commit history table shows the following commits:

Commit	Author	Message	Time
6e78475	Alvinbsi	second commit	2e984d0 - 1 minute ago
2e984d0	Alvinbsi	first commit	1 hour ago

The repository also includes a README file and a Jenkinsfile. The right sidebar shows the repository's activity, including 0 stars, 1 watching, and 0 forks. The 'Releases' and 'Packages' sections indicate no releases or packages have been published yet.

Dashboard > k8s_build_nov13 >

Status ✓ k8s_build_nov13 Add description

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

🔍 Pipeline Syntax

Stage View

Average stage times:
(Average full run time: ~24s)

	Pull Code From GitHub	Build the Docker image	Push the Docker image	Deploy on Kubernetes
#7 Nov 13 08:31 1 commit	1s	921ms	9s	887ms
#8 Nov 13 No Changes 2e984d0 second commit	1s	919ms	9s	622ms

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:

aws Services Search [Alt+S] Mumbai Varshika

Dashboard EC2 Global View Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity
- Reservations [New](#)

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

Load balancers (1/2)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

	Name	DNS name	State	VPC ID	Availability Zones	Type
☒	ae849e130b44403da85060735a26572	ae849e130b44403da85060735a26572-71710389.ap-south-1.elb.amazonaws.com	-	vpc-0b6f3578bb6f011...	ap-south-1a (aps1-az1)	classic
☐	api-alvin-k8s-local-sa6egj...	api-alvin-k8s-local-sa6egj...	✓ Active	vpc-0b6f3578bb6f011...	ap-south-1a (aps1-az1)	network

Load balancer: ae849e130b44403da85060735a26572

Scheme: Internet-facing Hosted zone: ZP97RAFLXTNZK Availability Zones: subnet-0dd42f3d2568a18ae ap-south-1a (aps1-az1)

✓ DNS name copied

ae849e130b44403da85060735a26572-71710389.ap-south-1.elb.amazonaws.com (A Record)

Not secure ae849e130b44403da85060735a26572-71710389.ap-south-1.elb.amazonaws.com

How to install Symf... How to install Symf... Google mod rewrite - How... nginx CORS on prox... How to enable Cros... enable cross-origin... cors - How do I add... Example Nginx conf...

Jane Doe

Avatar

- Designer
- London, UK
- ex@mail.com
- 1224435534

Skills

Adobe Photoshop 90%

Photography 80%

Work Experience

Front End Developer / w3schools.com

Jan 2015 - Current

Lorem ipsum dolor sit amet. Praesentium magnam consectetur vel in deserunt aspernatur est reprehenderit sunt hic. Nulla tempora soluta ea et odio, unde doloremque repellendus iure, iste.

Web Developer / something.com

Mar 2012 - Dec 2014

Consectetur adipisicing elit. Praesentium magnam consectetur vel in deserunt aspernatur est reprehenderit sunt hic. Nulla tempora soluta ea et odio, unde doloremque repellendus iure, iste.

Conclusion

This document outlines the end-to-end process of setting up a **Kubernetes project** using **Kops, Jenkins, Docker, and GitHub** on **AWS** infrastructure. By following these steps, you can automate the deployment of your Python application into a Kubernetes cluster seamlessly.

The process is broken down into clear stages:

1. **Setting Up Jenkins:** Using an Ubuntu EC2 instance to install Jenkins for continuous integration.
2. **Creating a GitHub Repository:** Managing your code in a version-controlled environment.
3. **Building a Jenkins Pipeline:** Automating code deployment using Jenkins to pull from GitHub and containerize with Docker.
4. **Containerizing with Docker:** Building and pushing Docker images to DockerHub for scalable deployments.
5. **Provisioning a Kubernetes Cluster:** Using Kops to create a fully managed Kubernetes cluster on AWS.
6. **Automating Kubernetes Deployments:** Extending the Jenkins pipeline to deploy containers onto your Kubernetes cluster.

By leveraging **CI/CD pipelines**, **containerization**, and **Kubernetes orchestration**, you've set up a highly efficient workflow that enables rapid development, testing, and production deployment. This infrastructure ensures scalability, resilience, and ease of management for your application.